GRAVIFIRE BASS PSI-26 Deep Research

GRAVIFIRE Deep Research Report

BASS / Burning and Suppression of Solids

Evidence Audit, Suppression Science, Literature Review, and Research Gap Map

Project: GRAVIFIRE - NASA International Space Apps Challenge 2026
Investigation: Burning and Suppression of Solids (BASS)
NASA PSI: PSI-26
NASA Center: Glenn Research Center
Flight Platform: International Space Station, Microgravity Science Glovebox
Historical PI: Paul V. Ferkul
Prepared: 1 October 2026

MEASURED DERIVED MODELED INTERPRETED OPEN / UNVERIFIED

Evidence standard used in this report
Claims are separated, where useful, into **Measured**, **Derived**, **Modeled**, **Interpreted**, and **Open/ Unverified**. Original BASS is not back-filled with BASS-II instrumentation or metadata. Planned test points, completed burns, archive files, and published analysis counts are treated as different quantities unless a source explicitly equates them.

Executive Summary

The original **Burning and Suppression of Solids (BASS)** investigation was an early long-duration ISS campaign focused on how solid fuels **burn, spread, persist, quench, and respond to active suppression in microgravity**. NASA used a small forced-flow duct inside the Microgravity Science Glovebox (MSG) and deliberately tested several fuel families and geometries, including SIBAL cotton-fiberglass fabric, Nomex, Ultem, PMMA/acrylic flat samples, wax-saturated fiberglass, PMMA spheres, and paraffin/Japan-wax candle configurations. S3 S5

The most distinctive scientific value of BASS is **intervention science**. BASS did not only observe flames. Investigators changed ventilation, reduced flow toward extinction, injected nitrogen suppressant, reused or preheated samples, and examined conditions where a suppressant jet could produce an unintended fluid-mechanical effect. The candle work documented a particularly important result: a nitrogen jet intended to suppress a flame could entrain oxygen-containing ambient air and **enhance combustion instead of simply extinguishing it**. S6

BASS also showed that microgravity flames can be exceptionally sensitive in the **0 to 5 cm/s ventilation region**. At very low flow, some flames became dim, blue, and stable rather than disappearing immediately. At modest low flow, continued heating of the solid could increase fuel vaporization and make the flame stronger with time. NASA explicitly noted the spacecraft-fire-safety implication: a small weak-looking flame might persist for a long time and later intensify if ventilation increases or another fuel is reached. S5

The investigation produced important model-validation evidence. SIBAL experiments supplied long-duration opposed- and concurrent-flow flame-spread data, including low-speed quenching. PMMA sphere experiments later supported transient spherical-fuel modeling and the development of follow-on growth/extinction research. Candle experiments provided a difficult benchmark involving radiation, soot, detailed chemistry, vitiated air, and suppression flow. S4 S7

BASS is therefore **not superseded by BASS-II**. BASS-II expanded systematic ignition/spread/extinction studies and atmospheric oxygen control, but original BASS retains distinctive evidence involving local suppressant jets, suppression failure through entrainment, candle flames, early spherical-fuel tests, broad exploratory materials, and dynamic ventilation interventions.

Most important conclusions for GRAVIFIRE

- **Do not reinvent:** basic SIBAL spread analysis, low-flow quenching, basic flame-front tracking, PMMA sphere thermal-history modeling, or the fact that nitrogen delivery can interact adversely with flow.
- **Potentially underused:** run-level suppression timelines, partial/failed suppression, dynamic overlay histories, repeated-sample thermal history, negative/near-limit cases, and archive-wide uncertainty metadata.
- **Most credible BASS opportunities:** suppression-event reconstruction, uncertainty-aware video measurements, temporal event ontologies, model applicability/discrepancy maps, contradiction-aware evidence synthesis, and evidence-state knowledge-gap mapping.
- **Do not flatten BASS into one ML table:** sheet spread, sphere regression, candle extinction, and suppression response are physically different targets.
- **Do not claim a final GRAVIFIRE direction from BASS alone:** these opportunities must later be compared with the other NASA fire experiment families.

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1. Investigation Identity

Field	Current evidence	Status
Investigation	Burning and Suppression of Solids	Verified
Acronym	BASS	Verified
NASA PSI	PSI-26	Verified from the official PSI endpoint and BASS-II related-investigation metadata
Completion	Complete	NASA PSI research-area listing
Research area	Combustion Science	Verified
Flight platform	International Space Station	Verified
Facility	Microgravity Science Glovebox	Verified
NASA center	Glenn Research Center	Verified
Historical PI	Paul V. Ferkul	Verified in NASA BASS documentation
Project Scientist / Co-I	Sandra L. Olson	Verified
Other documented Co-Is	James S. T'ien, Fumiaki Takahashi	Verified
Related investigation	BASS-II / PSI-25	Verified
PSI DOI	Not independently recoverable from the current indexed PSI-26 page	Open/Unverified
Current PSI version	Not independently recoverable	Open/Unverified
License	Not independently recoverable	Open/Unverified

The official PSI-26 route resolves at NASA PSI, but the current web index does not expose the same rich metadata that PSI-25 currently exposes. This report therefore **does not copy the BASS-II DOI, version, or license into BASS**. S1 S2

NASA's contemporaneous BASS quad chart identifies **Paul V. Ferkul** as PI, **Sandra Olson** as Project Scientist/Co-I, and James T'ien and Fumiaki Takahashi as Co-Is. S3

Timeline

The project has several meaningful time windows rather than one simple date:

- BASS planning and hardware development began before the main flight runs.
- Peer-reviewed and conference sources describe BASS operation aboard the ISS during **2012 and 2013**. S6
- NASA continued publishing BASS results and using BASS as a precursor/validation dataset in later years. S7

For GRAVIFIRE, individual run dates should be preserved when available instead of assigning a single publication-era date to all BASS evidence.

2. Why NASA Conducted BASS

2.1 NASA's stated spacecraft-fire problem

NASA's BASS objective was to help bridge the gap between **normal-gravity NASA-STD-6001 material screening**, short-duration reduced-gravity tests, and actual long-duration material flammability in microgravity. BASS also explicitly assessed an **inert gaseous extinguishing agent** similar to spacecraft fire suppression across different fuels, geometries, and flow configurations. S3

2.2 Why a 1g test is not automatically conservative

On Earth, heated gases rise. That buoyancy-driven flow changes oxygen delivery, flame shape, convective heat loss, and feedback from the flame to the solid. In orbital microgravity, natural convection is largely removed and the dominant transport can instead come from weak fan-driven spacecraft ventilation and diffusion. NASA therefore needed long-duration ISS data to determine whether materials that behave one way in a terrestrial flammability screen could behave differently under microgravity ventilation. S9

2.3 Why suppression had to be tested in flight

An extinguisher is not only a chemical or dilution source. A gaseous suppressant also creates a **momentum jet**. In microgravity, that jet can become one of the dominant local flow fields. The delivery can:

- dilute oxygen,
- move or stretch a flame,
- change convective heat transfer,
- entrain fresh ambient air,
- behave differently around partially obstructed or wake-stabilized flames.

NASA's BASS planning material explicitly anticipated that the correct point of suppressant application might not be obvious in microgravity. S3

2.4 Synthesis

BASS was simultaneously:

- **exploratory**, because it tested several fuels and geometries;
- **fire-safety oriented**, because suppression and ventilation were explicit objectives;
- **model-validation focused**, because microgravity removes buoyancy and simplifies the forced-flow boundary condition;
- **a precursor**, because later BASS-II and SoFIE work extended questions revealed by the original campaign. S13

3. Experimental Design

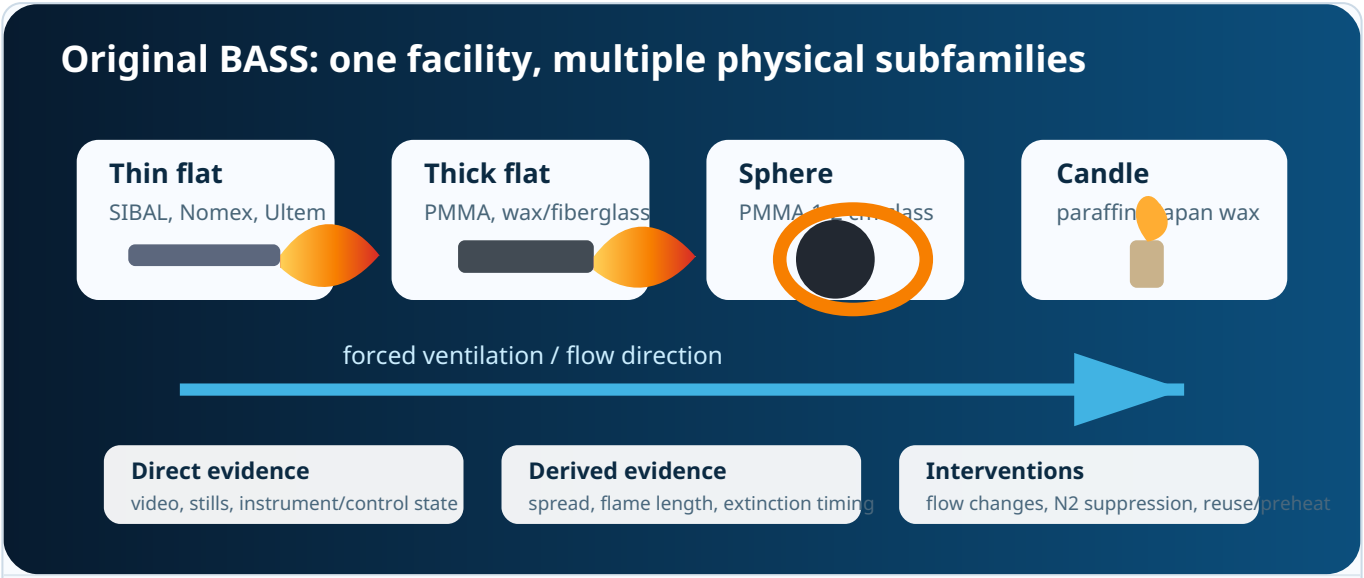


Figure 1. GRAVIFIRE schematic of the main original-BASS fuel/geometry families. Not an engineering drawing. It emphasizes why BASS must not be flattened into one homogeneous ML dataset.

3.1 Core facility and hardware

BASS used a small flow duct inside the ISS **Microgravity Science Glovebox**. NASA documentation describes slightly modified SPICE hardware with crew-controlled sample loading, ignition, airflow adjustment, nitrogen suppressant delivery, video downlink, selected high-resolution still imaging, and instrument/flow information associated with the run. S3 S10

3.2 Fuel families and geometries

Family	Materials / examples	Geometry	Why it mattered
Thin flat	SIBAL cotton-fiberglass, Nomex, Ultem	nominally long narrow strips	spread, limiting flame length, low-flow extinction
Thick flat	PMMA/acrylic, wax-saturated fiberglass	mm-scale slabs	thickness, preheat, thermal inertia
Sphere	PMMA	1 and 2 cm class spheres	axisymmetric geometry, thermal-history effects, possible reuse
Candle	paraffin and Japan wax in ceramic tube	wick/tube diffusion-flame geometry	wax combustion, low-flow flame structure, N2 suppression

The original BASS program is therefore **heterogeneous by design**. A sphere, sheet, and candle should not share a generic output variable called simply “flame spread” unless the specific measurement definition is preserved. S5 S6

3.3 Controlled and measured variables

Variable / parameter	Controlled?	Measured / derived?	Units	Notes
Fuel material	Yes	configuration	categorical	multiple fuel families
Geometry	Yes	configuration	sheet/sphere/ candle	physically non-interchangeable
Fuel thickness / size	Yes	physical sample property	mm, cm	key for thermal response
Forced airflow	Yes	instrumented	cm/s	particularly important from <1 to ~5 cm/s
Flow direction	Yes	configuration	opposed/ concurrent	fuel-family dependent
N2 suppressor flow	Yes	controlled	cc/min	documented around 500 cc/min in candle study
N2 nozzle geometry / location	configuration	documented for some studies	mm / position	affects entrainment
Oxygen fraction	configuration, limited	documented in particular tests	mole fraction	original BASS was not equivalent to BASS-II ambient-vitiation program
Ambient pressure	facility condition	contextual	atm	no systematic pressure axis
Ignition	crew controlled	timing observable	time	exact construction should be kept source-specific
Flame image / appearance	response	measured by camera	image/video	direct visual evidence
Flame front / spread	response	derived from imagery	distance/time	derived, not direct sensor value
Flame length / size	response	derived	length	geometry/camera dependent
Extinction / quench	response	observed	event/time/ condition	can be flow- or suppression-driven
Radiation	response	instrument signal in program documentation	relative/instrument units	absolute calibration should not be assumed without source support

3.4 Oxygen and pressure caveat

Original BASS did not provide the same systematic whole-MSG oxygen-control program that became important in BASS-II. Some BASS configurations used ordinary air or nitrogen-vitiated air, including the candle experiments at approximately 0.21 and 0.17 oxygen mole fraction. That does **not** make oxygen a universal BASS treatment axis. S6

BASS likewise did not systematically vary pressure. Pressure-dependent exploration-atmosphere fire behavior therefore cannot be learned from original BASS alone.

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4. Test Inventory and Count Audit

The public record contains **different count concepts** and they should not be silently merged.

4.1 Program-plan counts

Older NASA experiment-profile material describes a program with **41 fuel samples** and **89 planned test points**. The same broad profile family also contains BASS-II operational language, so these values should be treated as **program-plan figures requiring investigation-level provenance**, not automatically as a final BASS as-run count. S14

4.2 Recent NASA analyzed-burn count

A 2023 NASA fire-safety summary states that analysis of **59 BASS burn tests** produced data on heat flow, flame size, flow effects, and other parameters. This is a more direct modern NASA statement about analyzed BASS burns. S11

4.3 What this report treats as verified

Quantity	Value	How to interpret it
Program fuel-sample count reported in NASA profile	41	planning / program-level sample count, provenance caution
Program test points reported in older profile	89	planned test points, not automatically completed burns
BASS burn tests said to have been analyzed in 2023 NASA summary	59	modern NASA analysis count
Exact PSI-26 Experimental Table row count	not independently recovered	unresolved from current web-indexed PSI page
Exact completed physical runs	not fully reconstructed	open
Exact successful ignitions	not reconstructed	open
Exact raw-video count	not reconstructed	open

GRAVIFIRE rule: keep `planned_sample`, `planned_test_point`, `executed_run`, `successful_burn`, and `published_analysis_case` as separate entities.

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5. Public Dataset and Evidence Audit

NASA PSI lists BASS as a completed Combustion Science investigation and the official PSI-26 route is public. Current web indexing, however, does not expose the complete PSI-26 file manifest, version metadata, or all table rows in a way that can be independently audited here. S1 S2

Resource	Public evidence	What it contains	GRAVIFIRE usefulness
PSI-26 investigation repository	verified route	central NASA data archive	essential
Experimental/test database	documented as existing in NASA/PSI ecosystem	run/test metadata	essential, but exact current count unresolved
Raw video	verified by BASS publications and current NASA re-analysis	flame evolution, run dynamics, overlays	very high
High-resolution still images	verified	flame geometry and growth	high
NASA BASS quad chart	public PDF	objectives, team, experiment concept	essential context
BASS technical publication	public PDF	objectives, space/earth benefits, results	high
SIBAL analysis	NTRS public	opposed/concurrent spread, quench	very high
Thickness/preheat synthesis	NTRS public	multi-geometry BASS findings	very high
Candle/suppression conference paper	public metadata/full abstract	wax flames, vitiation, N2 entrainment	very high
Sphere model paper	peer reviewed	PMMA sphere precursor + model validation	high
Engineering/calibration files	likely archive content, not fully enumerated here	hardware/calibration	potentially very high
Machine-readable outcome ontology	not found	sustain/quench/suppression labels	opportunity
Archive-wide uncertainty schema	not found	measurement uncertainty	opportunity

Repository limitation

This is a **public-discoverability limitation**, not proof that NASA lacks the files. The report deliberately avoids guessing inaccessible filenames, file sizes, version numbers, or a Modeling Table that could not be directly verified.

6. Major Scientific Findings

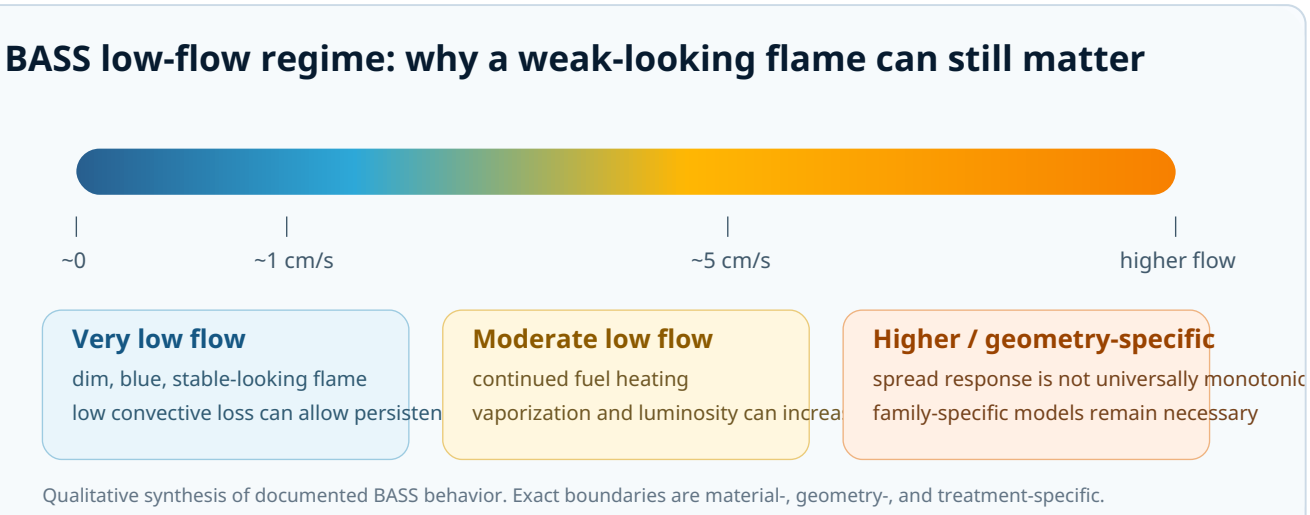


Figure 2. Qualitative evidence map based primarily on NASA BASS low-flow observations. It should not be interpreted as a universal flammability curve.

6.1 Extremely low ventilation can support weak but persistent flames

Measured/Observed: NASA's BASS synthesis reports especially strong sensitivity in the **0 to 5 cm/s** airflow region. Below about 1 cm/s, flames tended to become dim, blue, and stable. S5

Physical interpretation: low convective losses can allow a small flame to persist rather than extinguish immediately.

Spacecraft-fire implication: visual weakness does not automatically mean rapid self-extinction.

6.2 Modest low flow can strengthen a flame over time

At approximately 1 to 5 cm/s, NASA reports continued fuel heating, higher vaporization, and a stronger, more luminous flame as time progresses. Some small acrylic slab flames reportedly burned for **more than five minutes** before self-extinguishing. S5

This supports an important scenario:

```
small persistent flame -> continued fuel heating -> ventilation increase or new fuel -> stronger fire
```

This is an **experiment-supported concern**, not a universal probability model.

6.3 SIBAL opposed-flow spread was not simply monotonic with airflow

Derived from imagery: in opposed flow, the cotton-fiberglass flame quickly approached steady spread, and the spread rate was fastest at an **intermediate** airflow rather than increasing monotonically over the entire tested range. S4

This directly invalidates a simplistic GRAVIFIRE rule such as:

```
airflow up => flame spread always up
```

6.4 SIBAL concurrent-flow experiments produced rare long-duration transient data

In concurrent low-speed flow, flame size grew with time. At sufficiently low flow, the short sample could reach a limiting flame length and approximately steady spread. NASA emphasized that these ISS runs supplied detailed transient flame-growth data not readily obtainable in short-duration ground microgravity. S4

6.5 Low-flow quenching was directly observed

By decreasing flow toward roughly the 1 cm/s region, BASS observed quenching/extinction for SIBAL configurations. Later BASS/BASS-II work expanded the quenching boundary as a function of oxygen concentration. S4 S8

6.6 Thermal history matters for thick spherical fuels

The BASS PMMA-sphere precursor data later supported a transient spherical-fuel model. The follow-on paper emphasizes internal solid heating and shows why a thermally thick fuel cannot always be represented solely by its instantaneous external flow condition. S7

GRAVIFIRE implication: a reused/preheated sphere and a cold fresh sphere at the same nominal airflow may not represent the same physical state.

7. Fire Suppression Science

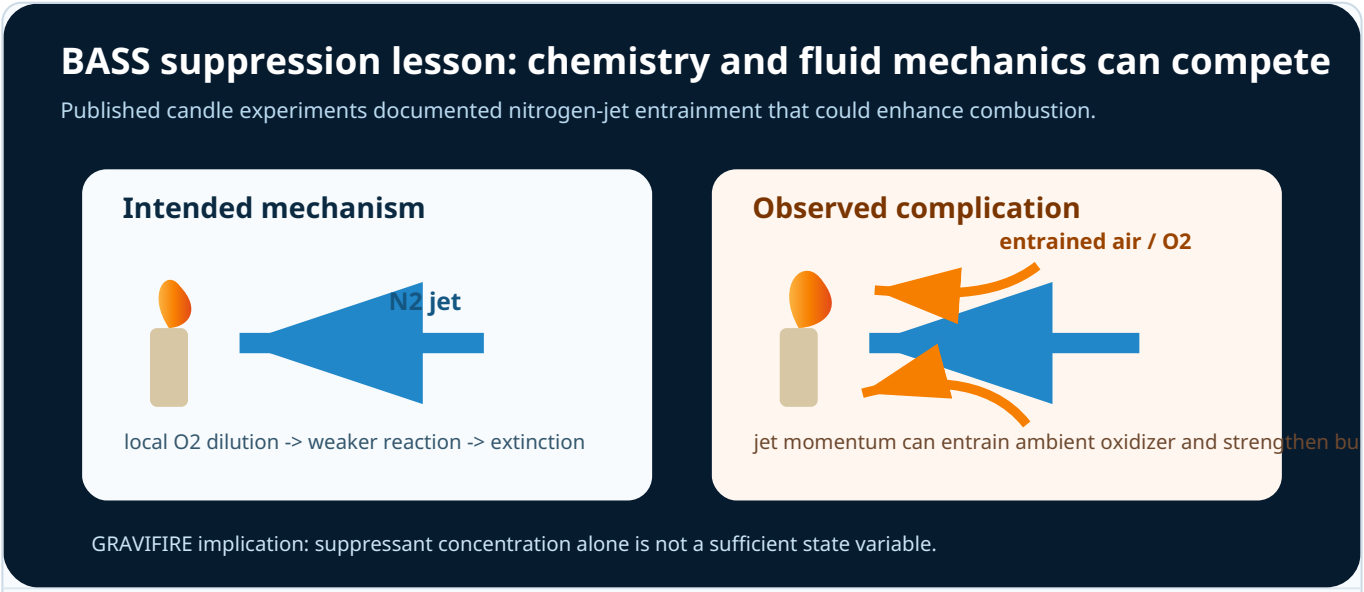


Figure 3. Schematic of the competing nitrogen-dilution and entrainment mechanisms documented in the BASS candle/suppression work. GRAVIFIRE schematic, not to scale.

Suppression is the original BASS feature most worth preserving as a separate scientific layer.

7.1 Suppression method

The principal documented BASS suppressor was **gaseous nitrogen**, delivered through a local nozzle. In the candle study, nitrogen was injected at approximately **500 cc/min** through a 3.1 mm ID tube upstream of the flame. S6

7.2 Intended mechanism

```
N2 jet -> local oxygen dilution -> reduced reaction support -> extinction
```

7.3 Observed complication: entrainment

The candle study documented the opposite fluid-mechanical effect:

```
N2 jet momentum -> entrains ambient oxygen-containing air -> increases oxidizer transport -> combustion enhanced
```

This is one of BASS's strongest direct contributions to spacecraft suppression understanding. A suppressant is not described by concentration alone; **delivery geometry and flow field matter**. S6

7.4 Ventilation shutdown

NASA interpreted BASS results as supporting the spacecraft fire-response principle of reducing forced ventilation, because selected flames extinguished rapidly when the primary airflow was removed. This should be classified as:

- **experimentally demonstrated:** airflow removal can extinguish specific BASS flames;
- **operationally relevant:** consistent with spacecraft ventilation-shutdown strategies;
- **not established by this review:** BASS alone created a new ISS emergency procedure.

7.5 Suppressant coverage limitation

No comprehensive original-BASS program comparing multiple suppressant technologies was found in the reviewed sources. BASS therefore supports important nitrogen-delivery physics but **does not support a universal spacecraft suppressant ranking**.

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8. Important Reports and Literature

Work	BASS evidence	Method	Main contribution	Limitation
Ferkul et al., 2012, *Flammability Aspects of a Cotton-Fiberglass Fabric…*	SIBAL	ISS experiment + image-derived spread	opposed/concurrent spread, enhanced microgravity flammability, quench	model fuel, small duct
Ferkul et al., 2013, *Thickness and Fuel Preheating Effects…*	flats, spheres, candles	multi-configuration synthesis	low-flow persistence, thermal history, suppression observations	conference presentation
Takahashi et al., 2013, *Burning Characteristics of Paraffin and Japan Wax Candle Flames…*	candle subset	ISS experiment + detailed-chemistry interpretation	low-flow wax flames, vitiation, N2 entrainment enhancement	conference-level publication; chemistry uses model assumptions
Zhao et al., 2017, *Concurrent flame growth, spread, and quenching…*	BASS + BASS-II SIBAL lineage	experiment + transient 3D modeling	quenching boundary and quantitative model comparison	combines two investigations
Endo et al., 2020, *Flame Growth Around a Spherical Solid Fuel…*	BASS sphere precursor	transient numerical model + experiment comparison	thermal history, sphere flame growth/ extinction	BASS sphere subset small
NASA Researcher's Guide to Combustion Science	program synthesis	NASA review	connects BASS to spacecraft fire-safety lineage	overview, not run-level data

8.1 SIBAL study

Title: *Flammability Aspects of a Cotton-Fiberglass Fabric in Opposed and Concurrent Airflow in Microgravity*
Authors: Paul V. Ferkul, Sandra L. Olson, Michael C. Johnston, James S. T'ien
NASA record: NTRS 20130012832. S4

Question: how does purely forced airflow affect flame appearance, transient growth, spread, and extinction in microgravity?

Contribution: the BASS data validated important theoretical predictions and provided rare long-duration data for transient model development.

8.2 Thickness and fuel preheating synthesis

Title: *Thickness and Fuel Preheating Effects on Material Flammability in Microgravity from the BASS Experiment*
Authors: Paul V. Ferkul, Sandra L. Olson, Fumiaki Takahashi, Makoto Endo, Michael C. Johnston, James S. T'ien
Venue: ASGSR 2013
NASA record: NTRS 20140011099. S5

It summarizes thin/thick flat samples, spheres, candles, airflow, preheat, nitrogen dilution/suppression, flame appearance, growth, and spread.

8.3 Candle and suppression study

Title: *Burning Characteristics of Paraffin and Japan Wax Candle Flames in a Low-Speed Oxidizing Stream in Microgravity*
Authors: Fumiaki Takahashi, Paul V. Ferkul, Sandra L. Olson, V. R. Katta
Venue: ASGSR/ISPS, 2013. S6

BASS configuration: wax in alumina tube, fiberglass wick, roughly 0 to 5 cm/s oxidizer flow, ordinary or nitrogen-vitiated air, opposed/concurrent orientation, N2 suppression attempt.

Key result: the nitrogen jet could entrain air and enhance combustion.

8.4 Sphere model-validation lineage

Title: *Flame Growth Around a Spherical Solid Fuel in Low Speed Forced Flow in Microgravity*
Authors: Makoto Endo, James S. T'ien, Paul V. Ferkul, Sandra L. Olson, Michael C. Johnston
Journal: *Fire Technology*, 56(1), 5-32
DOI: 10.1007/s10694-019-00848-2. S7

The paper explicitly identifies a BASS microgravity sphere experiment as a precursor used to assess experiment operation and validate the accompanying numerical model.

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9. Model Validation History

Model / approach	BASS cases	Inputs	Outputs	Validation finding	Remaining issue
Thin-fuel flame-spread theory	SIBAL opposed/ concurrent	flow, sample geometry, material properties	spread / flame behavior	important predicted trends supported	narrow fuel family
Transient concurrent flame-growth model	SIBAL	flow, oxygen in later combined studies, pyrolysis properties	transient flame growth, spread, quench	later BASS/BASS-II comparison quantitatively useful	family-specific
Spherical-fuel transient model	PMMA sphere precursor	flow, thermal properties, internal heating	flame growth, heat transfer, extinction	experimental trends agree with model	few precursor BASS sphere cases
Candle detailed-chemistry model	candle geometry	oxidizer flow, chemistry surrogate, thermal/radiative terms	flame structure / response	useful physical interpretation	surrogate chemistry not exact wax
Simple suppressant-dilution intuition	N2 jet	suppressant flow / oxygen	extinction	**insufficient alone**	entrainment can reverse expected effect

Model lesson for GRAVIFIRE

BASS does not justify one universal model. Each model has an **applicability domain**:

`fuel family + geometry + flow regime + measurement definition + assumptions`

GRAVIFIRE should expose that domain rather than hiding it behind a generic predictor.

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10. Image and Video Prior Art

BASS used real-time video and selected higher-resolution still imagery. The visual record has already supported extraction or interpretation of:

- flame-front position,
- spread rate,
- flame growth,
- flame length / size,
- extinction,
- sphere/flame geometry,
- flame appearance and color.

Current NASA-funded PSI re-analysis led by Ya-Ting Liao is processing BASS and BASS-II imagery using edge tracing, experiment-specific thresholds, flame-progression tracking, and extraction of real-time airflow information from overlaid video text. S12

Prior-art implication

The following is **not** a defensible novelty claim:

> "GRAVIFIRE is the first project to segment or track BASS flames from video."

What might still be scientifically useful

1. **Uncertainty-aware segmentation** - how much do flame measurements move when threshold, edge definition, scale, or frame quality changes?
2. **Suppression-event timelines** - synchronize N2 start, airflow, flame deflection, luminosity, quench, or enhancement.
3. **Repeated-sample history** - encode the thermal state of reused/preheated sphere tests.
4. **Evidence-linked events** - attach every derived event to exact frames, source files, and investigator descriptions.

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11. Machine Learning and AI Prior Art

Targeted searches were performed for BASS with terms including machine learning, neural network, Bayesian, surrogate modeling, active learning, computer vision, flame segmentation, image processing, uncertainty, and combined BASS/BASS-II analysis.

Technique	Evidence found specifically for BASS	Assessment
Classical image processing / edge tracking	Yes	established
Archive-scale BASS/BASS-II visual processing	Yes, current NASA project	active prior art
Neural-network BASS predictor	no direct evidence found in targeted search	**no evidence found**, not proof of nonexistence
Random forest / conventional supervised ML	no BASS-specific publication found	no evidence found
Bayesian predictive framework	none found	potentially open
Explicit uncertainty-aware prediction	no archive-wide framework found	underdeveloped
Active learning / next-experiment selection	none found	potentially open
Deep semantic segmentation	no BASS-specific peer-reviewed result found	but classical segmentation already occupies basic task
Cross-dataset BASS/BASS-II integration	yes	active prior art

Why generic ML is scientifically risky

A flat BASS feature matrix would mix physically incompatible targets:

- sheet flame-front spread,
- sphere regression / thermal response,
- candle extinction,
- nitrogen suppression response.

A model could obtain apparently good metrics while learning experiment-family identity rather than transferable physics. Any future ML study should use coherent subfamilies, explicit applicability domains, physics baselines, and leakage-resistant validation.

12. Contribution to Real Spacecraft Fire Safety

Experimentally demonstrated

- forced ventilation strongly changes microgravity solid-fuel behavior;

- weak low-flow flames can persist;
- low-flow quenching occurs in tested configurations;
- local N2 delivery can be affected by entrainment;
- microgravity behavior can differ markedly from the corresponding 1g geometry. S4 S5 S6

Model-supporting

BASS supplied relatively clean forced-flow microgravity evidence for validating flame-spread, transient growth, thick-fuel, sphere, and suppression-related models. NASA explicitly frames BASS results as contributing to computational models relevant to spacecraft fire detection and suppression. S10

Design-relevant

The data improve understanding of:

- ventilation control,
- suppressant-delivery geometry,
- limits of terrestrial material screening,
- weak or hidden flame persistence.

Operational-impact caution

This review did **not** find primary documentation proving that BASS alone directly:

- changed a named NASA-STD-6001 pass/fail threshold,
- certified or rejected a specific spacecraft material,
- installed a new detector or extinguisher,
- created a new ISS emergency procedure.

The documented impact is strongest at the level of scientific understanding, model support, fire-safety design relevance, and support for continued experiment development.

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13. Experimental Limitations

Scale

BASS used small samples and a small flow duct inside the MSG. It is not a spacecraft-scale fire experiment.

Gravity

ISS microgravity only. There is no controlled sequence through lunar and Martian gravity.

Pressure

No systematic BASS pressure axis was established in the sources reviewed.

Oxygen

Original BASS contains limited oxygen/vitiation cases but not the systematic atmospheric oxygen program that followed in BASS-II.

Materials

The material set is diverse enough for exploratory science, but still tiny compared with the material inventory of an operational spacecraft.

Suppressants

Nitrogen dominates the documented BASS suppression work. The experiment does not support general conclusions across all spacecraft extinguishing agents.

Diagnostics

No verified original-BASS full-field temperature, species, soot, or velocity field and no universally documented in-flight mass-loss measurement were found in this review.

Human-in-the-loop operations

Crew interaction with sample changes, ignition, flow adjustment, and suppressant delivery adds procedural variation that should be preserved in run provenance.

Thermal history

Repeated or preheated samples can carry state from earlier burning, violating an assumption of independent identically prepared trials.

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14. Dataset Limitations

Public discoverability

The PSI-26 route is public but the current indexed page did not expose a complete file manifest, current version, DOI/license metadata, or machine-readable table inventory to this audit.

Count ambiguity

Program plans, analyzed burn counts, executed runs, and usable scientific cases are different quantities. They should not be merged.

Dynamic treatment paths

A run can involve changing airflow, suppressant start, extinction, reignition, or reuse. A single static row may omit the most important causal sequence.

Video-embedded data

Current NASA work extracts experimental information from video overlays, demonstrating that some useful data are not conveniently represented as a normal table. S12

Missing universal outcome ontology

No verified archive-wide machine-readable label set was found for:

- failed ignition,
- sustained burning,
- steady spread,
- quench,
- successful N2 suppression,
- failed/partial suppression,
- suppressant-enhanced combustion,
- reignition.

Uncertainty metadata

No archive-wide schema was found that consistently attaches uncertainty to image-derived spread, flame length, extinction timing, airflow reconstruction, or suppressant timing.

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15. Research Gaps

Physics gaps

- exact low-flow transition boundaries across the complete material/geometry set;
- thermal-history effects beyond the limited thick-fuel/sphere cases;
- coupling between local suppressant jets and complex flame geometries.

Material gaps

- limited real spacecraft material diversity;
- no universal cross-material flammability relationship.

Environmental gaps

- no systematic pressure axis;
- no lunar/Mars partial-gravity sequence;
- limited exploration-atmosphere coverage.

Suppression gaps

- no comprehensive multi-agent comparison;
- incomplete run-level map of nozzle placement, jet momentum, flow, geometry, and outcome;
- lack of a validated universal suppression-response model.

Modeling gaps

- separate models exist, but no public BASS-wide model-applicability map was found;
- model failure/discrepancy cases are scattered across literature;
- no unified uncertainty propagation across measurement and model layers.

Data-usage gaps

- temporal event structure is not represented cleanly in normal experimental tables;
- video, test notes, model results, and literature claims are not exposed as one provenance chain;
- negative and partial-suppression cases may be especially informative but are hard to query.

AI / evidence-science gaps

- evidence-grounded scientific QA for BASS;
- automatic contradiction/context-difference detection;
- model-applicability reasoning;
- uncertainty-aware derived measurements;
- active next-experiment reasoning based on model disagreement rather than only sample density.

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16. DO NOT REINVENT THESE

Existing solution / finding	Maturity	Why GRAVIFIRE should not claim it as new
SIBAL opposed/concurrent spread analysis	strong experimental basis	directly reported from BASS
airflow strongly affects microgravity flames	core BASS result	foundational motivation/result
low-flow quenching	established in BASS lineage	directly observed and later modeled
basic flame-front / edge tracking	established	used in prior and current NASA analysis
PMMA sphere thermal-history modeling	peer-reviewed follow-on	BASS used as precursor validation
N2 jet can worsen combustion through entrainment	distinctive published finding	already documented in candle study
generic BASS+BASS-II visual meta-analysis	active NASA prior art	current NASA-funded project overlaps
ground tests do not automatically reproduce microgravity	pre-existing BASS motivation	not a new GRAVIFIRE discovery

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17. Potentially Underused BASS Data

17.1 Suppression attempts and failures

A high-value retrospective dataset would classify every recoverable suppression event by:

- fuel and geometry,
- flame state before intervention,
- background airflow,
- N2 flow and nozzle relation,
- onset time,
- flame deflection,
- luminosity/size change,
- success, failure, or temporary response.

Why it appears underused: the key entrainment effect is published, but an archive-wide suppression-response matrix was not found.

Risk: source metadata may not contain enough calibrated geometric detail.

17.2 Dynamic flow histories

Some BASS science is inherently a trajectory, not a point. Reconstructing `time -> control -> flame response` could recover scientific context lost in static rows.

Overlap warning: current NASA work is already extracting airflow histories from BASS/BASS-II imagery, so simple extraction alone is not novel. S12

17.3 Negative and near-limit cases

Failed ignition, weak persistent burning, partial quenching, suppression failure, and recovery after a control change are especially valuable for boundary science.

17.4 Repeated-sample thermal history

Sphere reuse creates an opportunity to encode material state over time rather than treating nominally identical conditions as independent.

17.5 Engineering/anomaly context

If PSI-26 contains detailed engineering logs or anomalies, linking them to runs could prevent downstream models from treating compromised or unusual conditions as normal samples.

18. BASS Research Gap Map

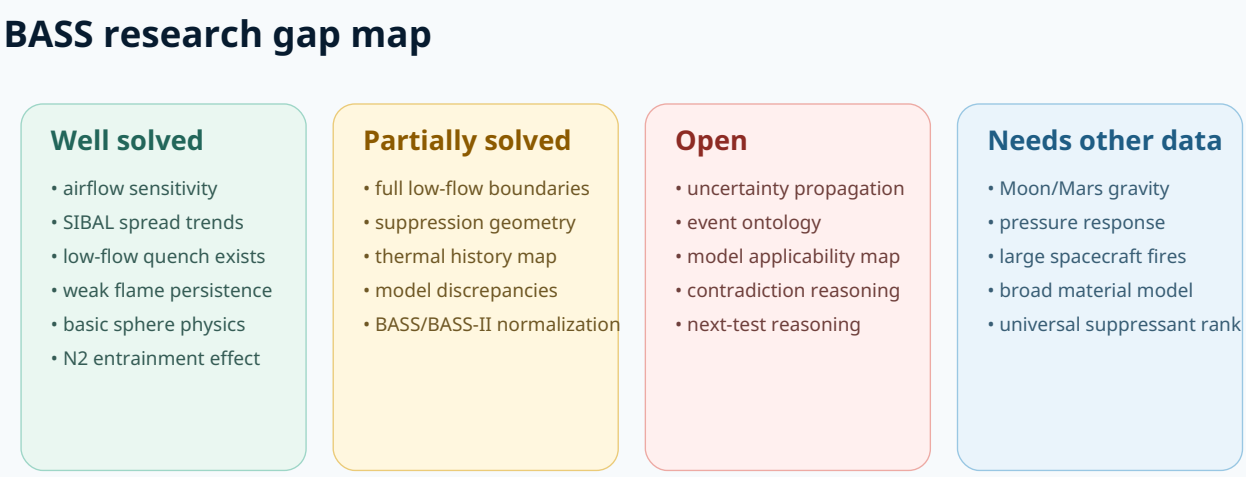


Figure 4. BASS research-gap map used to separate already mature questions from data-usage opportunities and questions requiring other experiment families.

WELL SOLVED

Within the tested domains:

- strong airflow dependence of microgravity solid flames;
- SIBAL opposed/concurrent spread trends;
- existence of low-flow quenching;
- long-duration transient flame growth;
- weak-looking flame persistence at very low flow;
- basic sphere growth/extinction physics and thermal-history importance;
- qualitative nitrogen-jet entrainment failure mode;
- conventional video-derived flame analysis.

PARTIALLY SOLVED

- quantitative generalization of SIBAL across broader material classes;
- complete mapping of low-flow transitions;
- suppression efficacy as a function of nozzle placement and jet momentum;
- thermal-history-dependent extinction beyond limited sphere work;
- ground-to-space material-test correlation;
- BASS-wide model validation and discrepancy synthesis;
- normalization with BASS-II while preserving family differences.

OPEN

- archive-wide uncertainty propagation;
- standardized suppression-event ontology;

- run-by-run model applicability map;
- model-disagreement atlas;
- contradiction-aware provenance reconciliation;
- systematic near-limit/failure-case analysis;
- scientifically constrained next-experiment recommendation.

DATA-RICH BUT POTENTIALLY UNDEREXPLOITED

- raw suppression videos;
- dynamic airflow overlays;
- suppressant-onset timing;
- failed/partial suppression cases;
- repeated sphere runs and preheated samples;
- unusual material cases;
- low-luminosity persistent flames;
- engineering and procedural metadata.

NOT SOLVABLE WITH BASS ALONE

- Moon-gravity prediction;
- Mars-gravity prediction;
- systematic pressure response;
- high-O2 exploration-atmosphere flammability;
- large-scale spacecraft fire growth;
- universal material flammability prediction;
- universal suppressant-agent ranking;
- smoke-detector performance;
- comprehensive toxic-product prediction.

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19. GRAVIFIRE Opportunity Matrix

Opportunity 1 - Suppression-event evidence reconstruction

Research question: Under what documented BASS conditions did nitrogen suppress, fail to suppress, or enhance combustion?

Data required: suppression videos, N2 timing/flow, airflow, sample geometry, test notes, source publications.

Prior work: Takahashi et al. establishes the entrainment-enhancement mechanism. S6

Potential new contribution: a run-level intervention-response map with frame-level evidence.

Method: temporal annotation + telemetry/overlay synchronization + provenance graph.

Validation: reproduce investigator-described cases and manually review ambiguous events.

Hackathon feasibility: High for a curated subset.

Main limitation: complete geometric/calibration metadata may be unavailable.

Novelty risk: Medium. The mechanism is known; the evidence reconstruction appears less developed.

Opportunity 2 - Uncertainty-aware video measurement

Research question: How uncertain are flame-front, flame-length, and extinction-time measurements extracted from BASS imagery?

New part: not segmentation itself, but **measurement + uncertainty + source quality**.

Method: vary threshold/edge definitions, frame-scale calibration, and temporal alignment; propagate measurement variation into spread/extinction quantities.

Validation: compare with published BASS measurements and check whether literature values fall within derived intervals.

Feasibility: High for selected runs.

Novelty risk: Medium. Threshold sensitivity is known, but explicit archive-level uncertainty remains underdeveloped.

Opportunity 3 - Temporal event ontology

Research question: Can each BASS run be represented as a physical sequence rather than a static row?

Example:

`ignition -> flame establishment -> spread -> airflow change -> N2 start -> deflection -> quench/enhancement`

Method: scientifically defined event schema with timestamps, evidence frames, confidence, and source references.

Validation: investigator reports and manual expert review.

Feasibility: High for a small evidence set.

Novelty risk: Medium-low. Tracking exists; a provenance-aware event ontology is more distinct.

Opportunity 4 - Model applicability and discrepancy atlas

Research question: Which published BASS models apply to which cases, and where do they agree or fail?

Output:

`model -> assumptions -> fuel family -> geometry -> valid regime -> test cases -> measured quantity -> agreement/disagreement`

Method: literature extraction + structured model cards + reproducible comparison where numeric outputs are available.

Validation: reproduce selected published comparisons.

Feasibility: High for curated literature.

Novelty risk: Low-medium. Individual comparisons exist, but no BASS-wide applicability layer was found.

Opportunity 5 - Contradiction-aware evidence synthesis

Research question: Can apparent source conflicts be classified as true contradiction, context difference, planning-vs-execution difference, or unresolved evidence?

Examples:

- 41 samples / 89 planned test points vs a modern NASA statement of 59 analyzed BASS burns;
- design flow ranges vs specific science-run ranges;
- program-level oxygen descriptions vs subset-specific vitiated-air conditions.

Method: claim extraction + provenance + context keys + manually reviewed benchmark set.

Feasibility: High if tightly scoped.

Novelty risk: Low based on the targeted BASS literature search.

Opportunity 6 - Evidence-state knowledge-gap map

Instead of only `tested / untested`, represent:

- directly measured,
- image-derived,
- model-supported,
- investigator-interpreted,
- suppression intervention,
- uncertain/conflicted,
- no evidence.

Feasibility: High.

Novelty risk: Medium, because current NASA meta-analysis overlaps with coverage mapping but not necessarily this epistemic-state framing.

Opportunity 7 - Model-disagreement next-experiment recommendation

Research question: Where would a future experiment best discriminate between scientifically plausible models or reduce the most consequential uncertainty?

Candidate score can combine:

- evidence sparsity,
- model disagreement,
- measurement uncertainty,
- feasible experimental domain.

Validation: retrospective holdout of known cases.

Feasibility: Medium.

Novelty risk: Low-medium. No BASS-specific active-learning framework was found, but future-experiment planning is an active NASA interest.

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20. BASS vs BASS-II

Dimension	Original BASS	BASS-II
PSI	PSI-26	PSI-25
Scientific character	exploratory flammability + suppression + model validation	deeper systematic ignition/spread/extinction studies
Platform	ISS MSG	ISS MSG
Hardware lineage	modified SPICE/BASS flow duct	reused/evolved BASS hardware lineage
Material emphasis	SIBAL, Nomex, Ultem, PMMA, wax/fiberglass, spheres, candles	more systematic PMMA families plus SIBAL and other configurations
Local N2 suppressant jet	**central distinctive feature**	not the defining BASS-II intervention
Systematic ambient O2 control	limited/subset-specific	substantially expanded
PMMA sphere precursor	**yes**	later related/follow-on science
Candle configurations	**yes**	not a principal BASS-II family
PMMA rod program	not central	major BASS-II family
Low-flow flame persistence	important	extended in more systematic parameter studies
Image/video evidence	yes	yes
Model-validation role	SIBAL, sphere, candle, low-flow suppression	extensive thin/intermediate PMMA, rods, SIBAL
Partial gravity	no	no
Pressure axis	no systematic program	no systematic program
Unique value now	dynamic suppression/intervention evidence	systematic spread/extinction parameter science

What BASS established

- spacecraft-relevant low-flow behavior can be counterintuitive;
- ventilation is a dominant microgravity control variable;
- small weak flames can persist;
- nitrogen delivery can interact adversely with local flow;
- several solid-fuel models can be meaningfully validated in long-duration microgravity.

Why BASS-II was needed

BASS revealed important behavior but left major questions under-parameterized. BASS-II expanded systematic study of ignition, flame spread, extinction, geometry, fuel dimensions, and ambient oxygen conditions while retaining the long-duration ISS forced-flow advantage. S15

What BASS remains uniquely useful for

- local N2 suppression events;
- suppression enhancement through entrainment;
- wax/candle configurations;
- broad exploratory materials;
- PMMA sphere precursor/thermal-history evidence;
- dynamic ventilation and intervention sequences.

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21. Preliminary BASS + BASS-II Synthesis

21.1 What the two together reveal

Together, the investigations trace a progression from **exploratory material/geometry/suppression behavior** to **more systematic environment and geometry parameterization**.

A defensible combined analysis is possible only if each measurement retains:

- investigation identity,
- fuel family,
- geometry,
- flow orientation,
- treatment history,
- measurement method,
- evidence status,
- uncertainty/provenance.

21.2 Variables that can potentially be harmonized

With source-specific calibration:

- material identity,
- dimensions,
- geometry class,
- opposed/concurrent orientation,
- physical airflow where verified,
- oxygen concentration where actually measured/controlled,
- timestamps,
- image-derived flame-front position,
- flame length,
- extinction/event status.

21.3 Variables that should not be directly merged

- BASS local N2 suppressor flow vs BASS-II working-volume oxygen dilution;
- sheet spread vs rod spread vs sphere regression vs candle flame length;

- static conditions vs deliberately ramped flow histories;
- fresh vs preheated/reused material state;
- uncalibrated instrument signals across configurations;
- planned settings vs as-run measured values.

21.4 Correct combined-data architecture

A scientifically defensible representation is hierarchical rather than a single generic ML table:

```
NASA solid-fuel fire evidence
|
+-- BASS / PSI-26
|   +-- thin flat / SIBAL
|   +-- thick flat
|   +-- PMMA sphere
|   +-- candle
|   +-- suppression interventions
|
+-- BASS-II / PSI-25
    +-- thin PMMA
    +-- intermediate PMMA
    +-- PMMA rods
    +-- SIBAL
    +-- other configurations
```

A unified query layer is scientifically reasonable. A universal predictor that ignores these family boundaries is not.

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22. Official NASA Imagery References

This report uses custom GRAVIFIRE scientific diagrams for portability and reproducibility. The following **official NASA-created BASS imagery** is recommended for visual evidence and presentation use:

1. **ISS035-E-014971 - BASS flame close-up, 6 April 2013.** NASA image, public-domain U.S. government work. NASA catalog image mirrored by Wikimedia Commons. S16
2. **BASS image sequence in NASA's 2022 ISS Imagery presentation.** The slide explicitly identifies a sequence from the second BASS run and explains its use for fire detection/suppression model development. S17
3. **NASA Researcher's Guide to Combustion Science.** Includes a BASS acrylic-sphere flame image, SIBAL flame sequences, and BASS suppression context. S9

> **Image-use note:** NASA-created imagery is generally public-domain in the United States unless otherwise noted, but NASA insignia/logos have separate usage restrictions. Each image should retain its NASA credit and photo ID/source where available.

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23. Final BASS Assessment for GRAVIFIRE

Original BASS has two different types of scientific value.

Physics value

It established real microgravity behavior involving low-speed flame persistence, spread, quenching, thermal history, and suppressant-flow interaction. Several of those phenomena are already well studied and should not be repackaged as new discoveries.

Evidence-science value

The archive appears to contain a more difficult and potentially valuable problem:

- dynamic interventions,
- partial/failed suppression,
- multiple physically distinct fuel families,
- video-embedded experimental state,
- reused/preheated samples,
- scattered model comparisons,
- uncertainty and provenance that are not naturally represented in a flat table.

For GRAVIFIRE, the most defensible BASS-specific direction is therefore **not another generic flame predictor**. The stronger opportunity is to make BASS evidence **temporally reconstructable, intervention-aware, uncertainty-aware, model-aware, and provenance-aware** while preserving what NASA actually measured versus what later researchers derived or modeled.

This is a **BASS-only conclusion**. It does not select the final GRAVIFIRE research contribution.

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24. Source List

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https://www.nasa.gov/physical-sciences-informatics-psi/psi-investigations-by-research-area/

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S4. Ferkul PV, Olson SL, Johnston MC, T'ien JS. Flammability Aspects of a Cotton-Fiberglass Fabric in Opposed and Concurrent Airflow in Microgravity. NASA NTRS 20130012832.
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S5. Ferkul PV, Olson SL, Takahashi F, Endo M, Johnston MC, T'ien JS. Thickness and Fuel Preheating Effects on Material Flammability in Microgravity from the BASS Experiment. NASA NTRS 20140011099.
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S6. Takahashi F, Ferkul PV, Olson SL, Katta VR. Burning Characteristics of Paraffin and Japan Wax Candle Flames in a Low-Speed Oxidizing Stream in Microgravity. 2013.

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https://doi.org/10.1007/s10694-019-00848-2

S8. Zhao X, Liao YT, Johnston MC, T'ien JS, Ferkul PV, Olson SL. Concurrent flame growth, spread, and quenching over composite fabric samples in low speed purely forced flow in microgravity. Proceedings of the Combustion Institute 36(2), 2971-2978.
https://doi.org/10.1016/j.proci.2016.06.028

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S11. NASA - Science in Space: Fire Safety in Space, 2023.
https://www.nasa.gov/missions/station/iss-research/science-in-space-week-of-sept-29-2023-fire-safety-in-space/

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S15. NASA PSI - BASS-II / PSI-25, related investigation metadata.
https://psi.nasa.gov/physci/repo/data/investigations/PSI-25

S16. NASA photo ISS035-E-014971, BASS experiment, mirrored by Wikimedia Commons.
https://commons.wikimedia.org/wiki/File:ISS-35_Burning_and_Suppression_of_Solids_(BASS)*experiment*(1).jpg

S17. NASA NTRS - International Space Station Imagery, 2022, BASS image sequence.
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